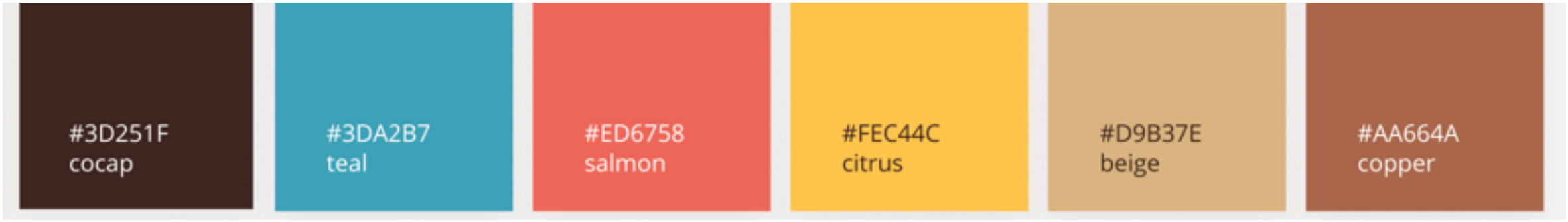


Gives a geometry on probability distributions

Guarantees continuity: small data changes $\rightarrow$ small change in distance



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper

So what?

- Why is having a proper metric useful/necessary?
- Mathematically:
 - **Gives a geometry on probability distributions.**
 - Enables tools from functional analysis & geometry (gradient flows, PDE links).
 - W_p isn't just a number — it equips probability distributions with a metric space structure.
- Machine Learning implications :
 - Compare datasets in a stable way (unlike KL or TV).
 - **Guarantees continuity: small data changes → small change in distance.**
 - Basis for algorithms: Wasserstein GAN, OT-based clustering, barycenters.

The Wasserstein Space $\mathbb{W}_p$